



Non-destructive detection of foreign contaminants in toast bread with near infrared spectroscopy and computer vision techniques

Jifan Yin^{1,2} · Saima Hameed^{1,2} · Lijuan Xie^{1,2} · Yibin Ying^{1,2}

Received: 9 April 2020 / Accepted: 26 August 2020 / Published online: 2 September 2020
© Springer Science+Business Media, LLC, part of Springer Nature 2020

Abstract

Non-destructive detection methods to identify food foreign contaminants meet the development of the food industry. Near infrared (NIR) spectroscopy and computer vision (CV) have been considerably intriguing due to the advantages of safety and rapidity. In this study, foreign contaminants, including metallic iron, polypropylene plastic, and hair in bread, were identified based on these two techniques. In NIR spectroscopy, the effectiveness of distance match and discriminant analysis methods combined with different spectral pretreatments was compared. It showed that the accuracy was 98%, 94%, 91%, and F-score was 0.97, 0.93, and 0.91 in the validation set for detecting foreign contaminants aforementioned when discriminant analysis combined with Savitzky-Golay smoothing was used. In CV, deep learning based on the modified U-net has applied to segment these contaminants on the surface of the bread; the accuracy of the test set was 95%, 93%, and 92%, respectively. Based on the results, it can be concluded that NIR spectroscopy and CV are both an operational way to detect foreign contaminants in bread. And these two techniques could be combined to apply in on-line detection further.

Keywords Bread · Foreign contaminants · Near-infrared spectroscopy · Deep learning · U-net

Introduction

Bread is one of the indispensable components of diets around the world, especially in Europe [1]. Its quality is of paramount importance during the process of bread production. Bread could be easily contaminated by foreign contaminants. Typical contaminants include metal, plastic, hair, etc., which pose a threat to human health as well as the company's fame [2]. Equipment, facilities, and cleaning tools are the source of foreign metal contaminants in packaged food. The primary source of plastic is flour packaging bags. While hair is brought in by the workers. Besides, food safety attracts the public's extensive attention, which is of vital importance for the food industry [3–5]. Therefore, it is

essential to find an effective method to detect foreign contaminants in bread. Manual sorting is still widely utilized in regions where labor rates remain low or industry is in small scale [6]. However, with the increment of the cost of labor and drawbacks to time-consuming with low efficiency and unverifiable accuracy, so automated techniques are needed. Currently, techniques, such as ferrous in foil metal detectors and balanced three-coil system to detect small metal particles [6], optical sorting systems, microwave imaging techniques, nuclear magnetic resonance imaging (NMRI), surface penetrating radar, electrical impedance and ultrasound are widely applied for food contaminant identification [7]. Especially, the optical sorting system, which provides a cosmetic enhancement to products by removal of blemished and discoloured product, are proper for detecting metal, plastic, and hair in bread.

Near infrared (NIR) spectroscopy technique has been emerging in recent years [8–10], which is straightforward, rapid, low-cost, and non-destructive [11–13]. The spectroscopy mainly reflects the characteristics of the measured object by offering information about the absorption of overtones and combinations of fundamental vibrations of hydrogen-containing groups [14, 15]. Currently, NIR spectroscopy has been adopted to detect defects in macadamia kernel

✉ Lijuan Xie
ljxie@zju.edu.cn

¹ College of Biosystems Engineering and Food Science, Zhejiang University, 866 Yuhangtang Road, Hangzhou 310058, People's Republic of China

² Key Laboratory of On-Site Processing Equipment for Agricultural Products, Ministry of Agriculture and Rural Affairs, Zhejiang University, Hangzhou 310058, People's Republic of China

quality [16] and flaws in hazelnut kernel [17]. The defects in green coffee also have been automatically detected based on NIR spectroscopy [18]. Besides, the insects in olives [19], chestnuts [20], rice [21] have been detected based on NIR spectroscopy.

Computer vision (CV) is increasingly applied in food inspection [22, 23] due to its objective, rapid, low-cost, non-contact, and non-invasive features [24–26], which could overcome subjective differences and visual fatigue. The principle is converting the image signal containing visual attribute information into digital signals and then processing it by the algorithm to simulate human discriminant criteria to understand and recognize images [27]. Recently, researchers exert great appeal to deep learning, which is highly practical. The remarkable and unique advantage is it could identify data by training set instead of being affected by hand-made parameters set by users. As for a powerful method for food control, it has been applied to the agricultural field extensively. By deep learning, foreign impurity in kernels [28], and lettuce [29] have been identified, the browning stages of bread crust [30] were recognized, fish freshness diagnosis [31] was achieved, and plant disease was identified [32].

In this paper, the feasibility of detecting foreign contaminants in bread is assessed based on NIR combined with CV techniques. For NIR spectroscopy, two modeling methods were compared, including discriminant analysis (DA) and distance match (DM) coupled with different spectra pre-treatments. For CV, the modified fully convolutional neural network (FCN) structures were utilized to segment various foreign contaminants on the surface of the bread, and this is the first time that FCN is applied to the detection of foreign contaminants in food.

Materials and methods

Materials

Bread, mainly made of wheat flour, water, sugar, and milk, was purchased in the mart (Hangzhou, China). The type of bread is toast. A total of 920 cubes was obtained and used in this study. All samples were cut into about 25 mm × 20 mm × 10 mm (length × width × thick) cubes in the laboratory. Metallic iron, polypropylene plastic, and hair were prepared as foreign contaminants. The sizes of metallic iron include 4 mm × 1 mm (length × width) and 2 mm × 1 mm. The sizes of polypropylene plastic include 10 mm × 2 mm (length × width), 8 mm × 2 mm, and 6 mm × 2 mm. For hair, the size of it is about 70 μm × 8 mm (diameter × length) and 70 μm × 6 mm. For detecting foreign contaminants for NIR spectroscopy collection, 320 cubes were used, 80 of which were pure bread cubes, the rest of which were mixed with metal, plastic, and hair with

80 cubes for each group. These foreign contaminants are inserted in the bread about 3 mm deep. For CV, different foreign contaminants were stuck to 400 cubes and 200 pure cubes.

Near infrared spectroscopy system

Near infrared spectroscopy system mainly consists of the infrared spectrometer to acquire spectra information and a computer to save and process data. Generally, a NIR spectrometer is composed of a radiation source, an interferometer, a sample holder, and a detector for the absorptivity measurement [33]. In the study, the instrument employed was the Fourier transform near-infrared spectrometer (Thermo Nicolet, USA) equipped with a 50 W quartz halogen lamp as the radiation source and an InGaAs detector. The spectra have been collected in reflectance mode. The ambient temperature and relative humidity were 25 ± 2 °C and 60%, respectively. Instrument parameters, including scan times and resolution, were set up to 32 and 1 cm⁻¹, respectively. Eventually, raw spectra were exported using OMNIC 8.1 (Thermo Nicolet, USA). For each sample, three spectra were collected, and an averaged spectrum was obtained. For collecting spectra of bread with foreign contaminants, the region where contaminants are beneath it was chosen.

Computer vision system

The computer vision system consists of a CMOS camera (Lucid Phoenix PHX050S-PC, Canada), the lighting source comprising two white LED stripe, and a plate for samples. Focal length and gain are set as 50 mm and 26 dB, respectively. The resolution of images is 2448 × 2048 pixels and frame per second of 24. The CMOS camera was placed 30 cm above the scene. Images were acquired by placing bread manually to orientate the side with foreign contaminants towards the camera. The configuration of this system is shown in Fig. 1.

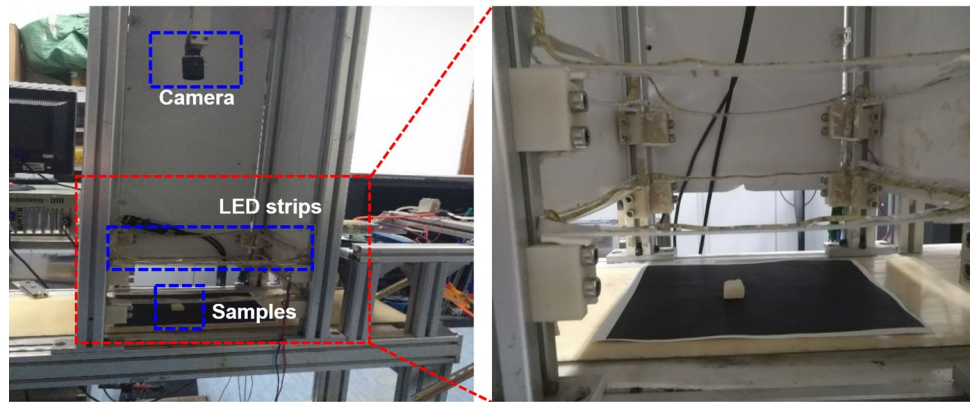
Foreign contaminants identification by near infrared spectroscopy

Spectroscopy pre-processing

Besides useful information, spectra also contain a lot of background noise and irrelevant information. Therefore, it is necessary to pre-process the spectra to weaken the influence of various non-target factors, retain valid data, reduce the complexity of the model, and improve the robustness of the model [34].

For noise reduction, the Savitzky-Golay is a universal smoothing method, which eliminates noise with the lowest useful information distortion [35]. The principle is based on

Fig. 1 The configuration of the imaging system



simplified least squares proposed by Savitzky and Golay first in 1964 [36]. Derivative methods are mainly applied for baseline correction and solving the problem overlapped and hidden bands.

In the study, the optimum one was found by comparing with these pre-processing methods.

Principle component analysis

Principle component analysis (PCA) was applied to reduce the data dimension by an orthogonal transformation, which transformed correlated variables into variables that are linear uncorrelated [37]. During the process, data were compressed by eliminating noise and other redundant information.

Chemometrics for modeling

Chemometrics signifies the application of statistical, mathematical methods to extract valid information from chemical and physical data. In the study, qualitative analysis was applied to detect whether the bread was contaminated. Discriminant analysis (DA) and distance match (DM) were used to discriminate against the contaminants in the study.

Discriminant analysis is to extract classification information by project samples into the optimal discriminant vector space, ensuring the most substantial inter-class distance and the smallest intra-class distance in the new subspace. The most commonly distance calculated are Euclidean distance (ED) and the Mahalanobis distance (MD) [38]. And the latter has a broader scope of application due to considering correlation among different attributes and scale-invariant. It is expressed as

$$u = (u_1, u_2, u_3, \dots, u_p)^T$$

$$x = (x_1, x_2, x_3, \dots, x_p)^T \quad (1)$$

$$D_M(X) = \sqrt{(x - u)^T \Sigma^{-1}(x - u)}$$

Here, u is the mean matrix, x is the covariance matrix, $D_M(X)$ is the Mahalanobis distance.

A distance match is suitable for identifying contaminants with a slight difference. While using this method to classify unknown samples, the average spectrum and standard deviation spectrum are obtained. The residual spectrum is obtained by subtracting the average spectrum from the spectrum of unknown samples, later, which is divided by the standard deviation spectrum. The result is called the matching value. When it is the lowest in a specific category, it represents this spectrum matches this category best. The above description can be seen in the Eq. (2)

$$Z_{ei} = e_i / S_e = (y_i - \hat{y}_i) / S_e \quad (2)$$

z_{ei} is the matching value, e_i is the residual spectrum, y_i is the unknown spectrum, $\hat{y}_i$ is the average spectrum, and S_e is the standard deviation spectrum.

In this study, the above chemometrics methods were operated by TQ Analyst 8.0 (Thermo Nicolet, USA) to explore which one of these methods is the most suitable for discriminating the contaminants.

Foreign contaminants identification by computer vision

Data

A total of 600 images with and without various contaminants were obtained for deep learning. Each image consists of ten bread cubes. Hence, there are 6000 toast bread cubes for this experiment. Among these, 200 images with contaminants and 100 images of pure bread were used for the training set, 80 images with contaminants and 40 images of pure bread were used for the validation set, and 120 images with contaminants and 60 images of pure bread were used

for the test set. For the training set, contaminants were annotated manually. Images were resized into 512×512 pixels to accommodate convolution calculation better.

Deep learning algorithm

Conventional segmentation algorithms are not proper in this situation. Taking the gray-level algorithm, for instance, indeed, it could segment contaminants based on their color diversity. However, it only acquires high accuracy when it is significantly distinct among contaminants and between them and bread. Also, the threshold value is set by the operator, and the environmental situation has adverse effects on its stability [39]. The goal is to avoid these harmful elements and to extract features automatically by deep learning. Convolutional neural networks (CNN), as the name suggests, based on convolutional calculations that mainly focus on classifying images instead of segmentation and the output, which is a probability value [40]. While FCN, unlike CNN, which requires a full connection layer, finally outputs an image [41]. Ronneberger et al. [42] modified and extended FCN, which is named “U-net”. The most appealing advantages are trivial training data available is required, and there are several feature channels in the upsampling, which beneficially devotes to the high resolution of images. For the detection of foreign contaminants, the U-net is the optimal method for this deep learning. Considering the complexity and irregular structures of objects, a modified U-net version, proposed by Dian Rong [28], was applied in the experiment. The architecture of the modified U-net is shown in Fig. 2. The bread image segmentation model has 16 layers of the convolution layer, 15 layers of batch normalization layer, 6 layers of maximum pooling layer, 12 layers of sampling layer, 15 layers of the ReLU function activation layer, and 1 layer of the sigmoid function. Two sides are for contracting and expanding, respectively. For subsampling, seven layers of convolution were set, each followed by batch normalization, the rectified linear unit (ReLU) layer, and a 2×2 max pooling operation with stride 2. ReLU is the function of Eq. (3).

$$f(x) = \max(0, x) \quad (3)$$

where x is the input data.

In subsampling, channel features were doubled from 16 to 512. For upsampling, the transposed convolution calculation was used, and eight layers of convolution were set, each followed by batch normalization and the ReLU. Here, six concatenations were shared in the network between the subsampling and upsampling to transfer and share more multiscale detailed information on bread and foreign contaminants. For upsampling, channel features were reduced from 512 to 16. After the upsampling, sigmoid was used to obtain a data map. Sigmoid is the function of Eq. (2).

$$S(x) = \frac{1}{1 + e^{-x}} \quad (4)$$

where x is the input data.

Performance index

In general, a confusion matrix is used for evaluating the classifier's performance. In this study, accuracy was used for assessing whether pre-processing could improve the quality of spectra, and deep learning could segment regions with contaminants. Precision (P), Recall (R), and F-score (F) were used to assessing the ability to detect different kinds of contaminants. Images with contaminants were defined as a True Positive (TP) class and images with pure bread as True Negative (Tn) class in the prediction models. Hence, False Positive (FP) and False Negative (FN) represented images with pure bread identified as images with contaminants and images with pure bread identified as images with contaminants, respectively. The equations are shown below.

$$R = \frac{TP}{TP + FN} \quad (5)$$

$$P = \frac{TP}{TP + FP} \quad (6)$$

$$F = \frac{2P \times R}{P + R} \quad (7)$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

Results and discussion

Near infrared spectroscopy analysis of pure substances

The original NIR spectra are presented in Fig. 3 between $12,500$ and 4000 cm^{-1} of pure materials, including bread, metallic iron, polypropylene plastic, and hair. It showed that there is a remarkable discrepancy among them.

Bread is mainly composed of water, protein, fat, and carbohydrate. From the spectrum of pure bread, it could be observed that there are strong absorption peaks near 8400 cm^{-1} , 6870 cm^{-1} , and 5100 cm^{-1} . The strong absorption peaks of 5100 cm^{-1} are the combined frequency peaks of the anti-symmetric stretching and bending vibration of water molecules [43]. The other peaks are mainly related to the first and second frequency doubling of the C–H group, O–H group, C=O group, and the combined frequency absorption bands of stretching and bending vibration [43].

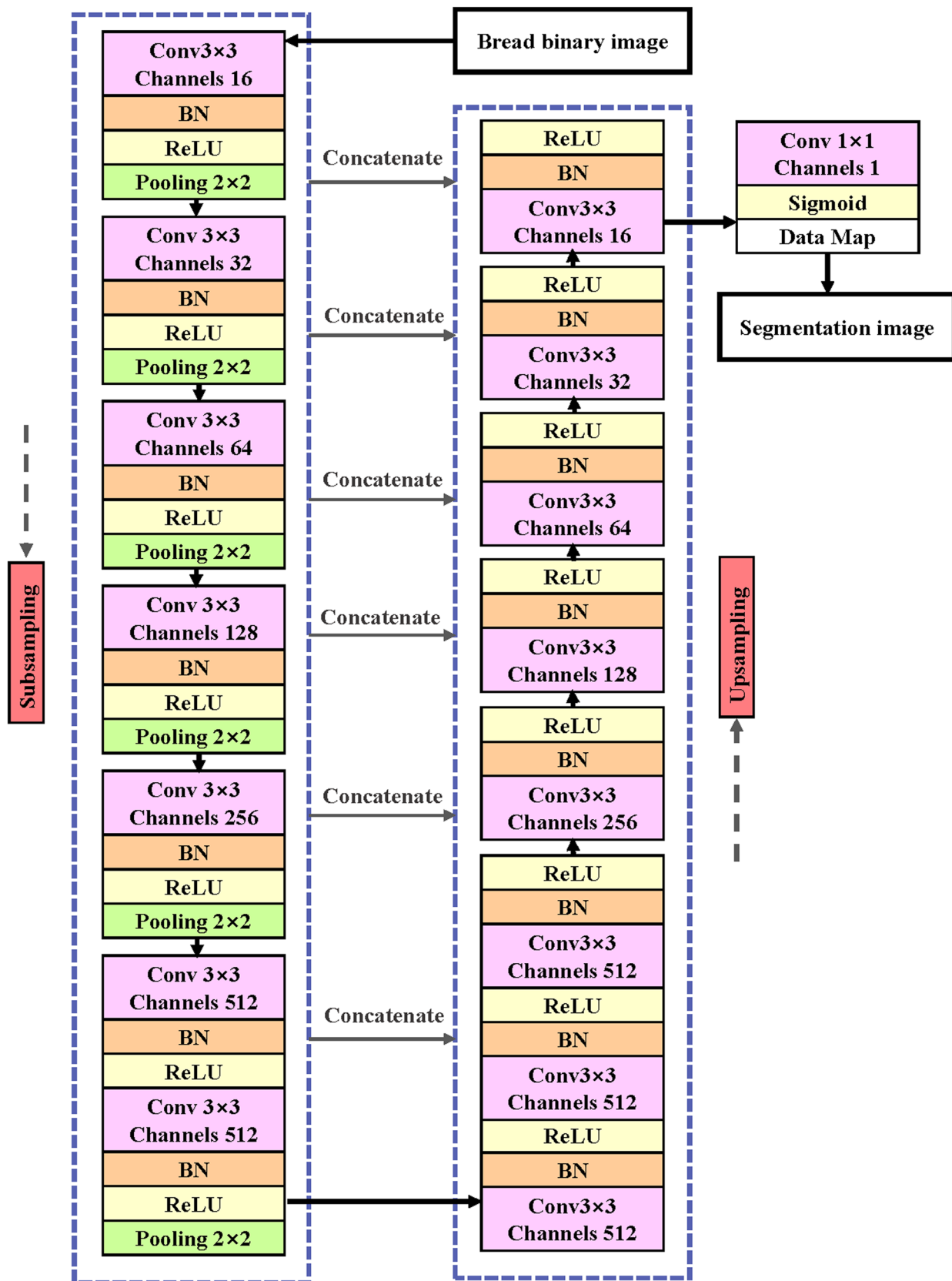


Fig. 2 The modified U-net for segmentation of bread images with foreign contaminants

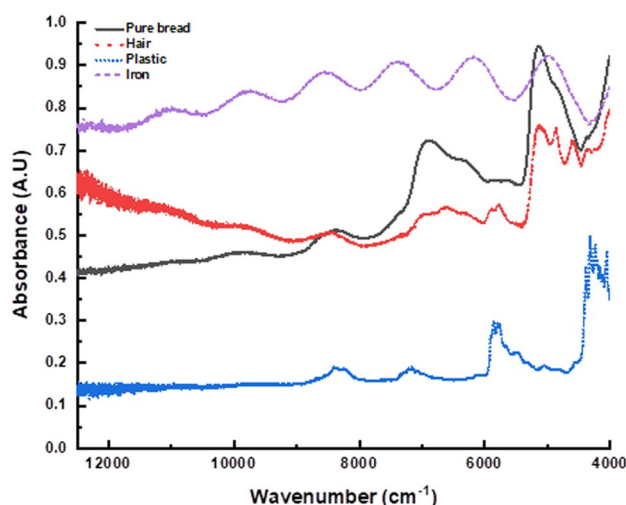


Fig. 3 Near infrared spectra of pure bread, hair, plastic and iron

Around the wavenumbers at 8400 cm^{-1} , 7200 cm^{-1} , 5800 cm^{-1} , and 4300 cm^{-1} , the NIR spectrum of polypropylene plastic showed strong absorption peaks, which denote the structure of polypropylene molecule. The leading information is related to the characteristics of the first and second frequency doubling and combined frequency of stretching and bending vibration of the CH_3 group, CH_2 group, and CH group [43].

Hair mainly consists of keratin; therefore, its near infrared spectrum reflected the information of the protein. The absorption peaks around 6660 cm^{-1} , 5800 cm^{-1} , 4860 cm^{-1} , and 4560 cm^{-1} are related to the first frequency doubling of N–H group stretching vibration, the first frequency doubling of C–H group stretching vibration, the combined frequency of N–H group stretching bending and the second frequency doubling of bending [43].

The near infrared spectrum of metallic iron oscillates uniformly, which is associated with the optical property of high reflectivity.

Principle component analysis

Due to the little difference between all samples' spectra and the huge amount of data processing, principle component analysis (PCA) was applied to assess its performance, which also further determines the possibility of detection by NIR spectroscopy technique. The number of principle components (PCs) was set at 2 in this study for the cumulative contribution rate of the first two PCs up to 98.20%. The proportion of the first and second PCs was 56.17% and 42.03%, respectively. Figure 4 showed the scores of the first two components of the spectra (PC1 as X-axis and PC2 as Y-axis), in which each sample was performed individually. It can be seen obviously that all samples are divided into

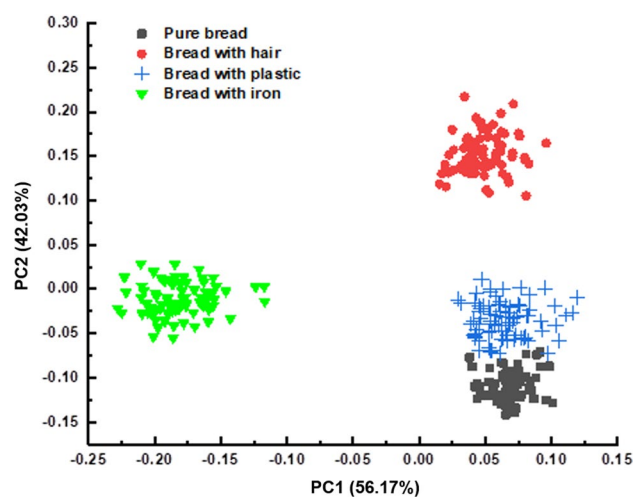


Fig. 4 Scores of the first second principle components

four clusters based on their first two orthogonal PCs, representing pure bread, bread with metallic iron, polypropylene plastic, and hair without any overlapping. Therefore, it can be concluded that bread with foreign contaminants could be separated from pure bread based on the spectra.

Chemometrics for modeling

Distance match

The effect of pre-processing methods on DM model accuracy is shown in Table 1. It reveals that the poor performance of the model using raw spectra with 49% accuracy can be enhanced by some pre-processing methods. Compared with the original spectra, the first and second derivatives can significantly improve the accuracy of the model. Savitzky-Golay smoothing does not affect the model accuracy of the original spectra or that with the first derivative. But it is worth noting that Savitzky-Golay smoothing has a synergistic effect with the second derivative to improve the accuracy of the model, the combination of which has the highest accuracy up to 91% among all the pre-processing methods aforementioned.

For the DM model with Savitzky-Golay smoothing and the second derivative, evaluation of DM with in terms of confusion matrix was analyzed, and Table 2 was the precision, recall, accuracy, and F-score of this model. In the calibration set, it turned out that the accuracy for metallic iron, polypropylene plastic, and hair discrimination was 98%, 95%, and 90%, respectively. Besides, F-score was in accordance with the accuracy, which was 0.98, 0.94, and 0.91 for metallic iron, polypropylene plastic, and hair, respectively. It was demonstrated that the DM model performed well in detecting foreign contaminants in bread, and the ability of

Table 1 The accuracy (%) of DM and DA with various pre-processing methods

Pre-processing methods	Raw spectra	SG smoothing	The first derivative	The first derivative + SG Smoothing	The second derivative	The second derivative + SG Smoothing
DM	49	49	90	90	88	91
DA	91	92	61	61	61	63

Table 2 Comparison of various classification models in terms of precision, recall, accuracy and F-score

		Calibration set				Validation set			
		Precision	Recall	F-score	Accuracy (%)	Precision	Recall	F-score	Accuracy (%)
DM*	Metal	1.0	0.95	0.98	98	1.0	0.93	0.96	96
	Plastic	0.97	0.91	0.94	95	0.97	0.89	0.92	92
	Hair	0.94	0.89	0.91	90	0.93	0.87	0.89	89
DA*	Metal	1.0	0.97	0.99	99	1.0	0.95	0.97	97
	Plastic	0.98	0.92	0.95	95	0.96	0.90	0.93	93
	Hair	0.96	0.90	0.93	92	0.94	0.88	0.91	91

DM DM model with the second derivative + SG Smoothing, DA DA model with the SG Smoothing

metal identification was the best, plastic identification is the second, hair identification is the lowest (90%). The result was related to the algorithm and the similarity of the NIR spectra of these substances seen in Fig. 3. In the validation set, it turned out that the accuracy for metallic iron, polypropylene plastic, and hair discrimination was 96%, 92%, and 89%, respectively. Besides, F-score was in accordance with the accuracy, which was 0.96, 0.92, and 0.89 for metallic iron, polypropylene plastic, and hair, respectively. From the result, it revealed the excellent generalization ability of this model.

Discriminant analysis

The comparative results of the accuracy of the DA model with different pre-processing methods could be seen in Table 1. The accuracy of the DA model using the original spectra is 91%, which performed significantly better than the DM model. Savitzky-Golay smoothing has a slight positive effect on modeling, which made the accuracy up to 92%, higher than the highest accuracy of DM. Meanwhile, it should be noted that the accuracy of the model with the first derivative is unchanged after combining with Savitzky-Golay smoothing while there is a synergistic effect combined with Savitzky-Golay smoothing and the second derivative, the conclusion of which is consistent with that of DM model. However, in this method, the accuracy with models using these pre-processing methods declined remarkably except Savitzky-Golay smoothing. It was demonstrated that improper pre-processing methods could reduce the quality of spectra due to the effective information removed.

For the DA model with Savitzky-Golay smoothing, evaluation of DA in terms of confusion matrix was analyzed, and Table 2 was the precision, recall, accuracy, and F-score of this model. In the calibration set, the accuracy of metallic iron, polypropylene plastic, and hair discrimination was 99%, 95%, and 92%, respectively. Additionally, F-score was in accordance with the accuracy, which was 0.99, 0.95, and 0.93 for metallic iron, polypropylene plastic, and hair, respectively. It was concluded that the DA model could detect foreign contaminants in bread. In the validation set, the accuracy of metallic iron, polypropylene plastic, and hair discrimination was 97%, 93%, and 91%, respectively. Additionally, F-score was in accordance with the accuracy, which was 0.97, 0.93, and 0.91 for metallic iron, polypropylene plastic, and hair, respectively. From the result, the model manifested the excellent generalization ability.

Compared with the DM model, it could be found that both two models could get high F-score and accuracy for detecting foreign contaminants. Also, both models for identifying metallic iron in bread performed the best; polypropylene plastic performed the second. For hair, it also got the accuracy and F-score in the validation set more than 89% and 0.89. It was worth noting that the DA model had slightly improved the ability compared with the DM model in the detection of these foreign contaminants; therefore, the DA model was recognized as the best model for the detection of such impurities.

FCNs for image segmentation

As shown in Fig. 5, it is one of the results of the detection of foreign contaminants based on the modified U-net

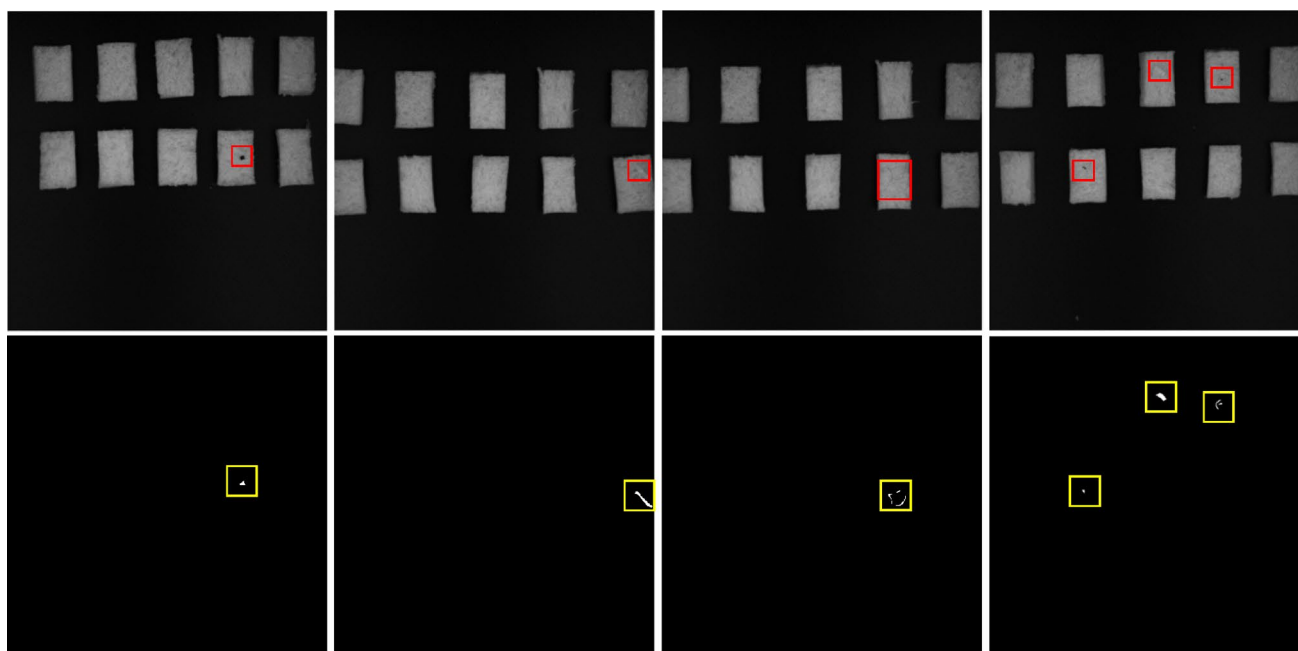


Fig. 5 Example of foreign objects detection in bread images. The binary image of bread with metal, plastic, hair and all aforementioned, respectively (red square, first row). Results of detection of metal, plastic, hair and all aforementioned, respectively (yellow square, second row)

deep learning algorithm. The binary images were in the first row, and there was some little impurity on the surface of bread circled by red squares, metal, plastic, hair, and mixture above from left to right, respectively. In the second row, it is shown that foreign contaminants were detected by the modified U-net, circled by yellow squares. The top and bottom of each column is a one-to-one match. In Fig. 5, it can be seen clearly that all foreign contaminants were segmented successfully, which demonstrated that the modified U-net is proper for the detection of foreign contaminants. During this process, the modified U-net realized the automatic segmentation of foreign objects without setting the threshold manually. However, through careful observation of segmented images, it can be found that although foreign matters were identified, they are not entirely segmented. The reasons, mainly include the following two points. On the one hand, it is the supervised training, which means foreign contaminants of bread in the train set was cutout manually in advance, and it could lead to some edge deviation. The consequence of that is not all the pixels of contaminants are trained. On the other hand, the instinct property of foreign contaminants and also bread corresponds with it, which is to be elaborated in detail in the next paragraph, which is also the main reason for the accuracy of detection.

Detection accuracy by U-net deep learning

The evaluation of the modified U-net for detecting foreign contaminants in terms of confusion matrix of the training

set, validation set, and test set was illustrated in Table 3, and the precision, recall, F-score, and accuracy of the training set, validation set, and test set as shown in Table 4. For the training set, the accuracy and F-score were 100% and 1.0. For the validation set, the two indexes were 94% and 0.95, respectively. For the test set, the two indexes were 93% and 0.94, respectively. The result of the test set has been analyzed in detail in the next paragraph.

There are 180 images for the test set. In each image, ten cube bread was placed with different foreign contaminants placed randomly. According to the results, it revealed that pure bread cubes were not segmented, which meant that bread itself was not recognized as foreign contaminants. The accuracy of foreign contaminants is illustrated in Table 5, from which it can be seen that deep learning based on the modified U-net performed well in segmentation for foreign contaminants. The accuracy was up to 92% as a whole. For metallic iron, the correct segmentation accuracy was 95%, which attributed to the full pixel information of metallic iron and the same to the reason that the accuracy of detecting polypropylene plastic is 93%, which also was a satisfying result. While for hair, there was a slight inaccuracy, which was mainly due to the size of the hair. The diameter of hair generally was 70 μm ; therefore, during the process, it would be recognized as a weak signal.

Deep learning, based on the modified U-net, could obtain desirable results for detecting foreign contaminants on the surface of the bread. There are rare research applying FCN to detection of foreign contaminants in food, and it was utilized

Table 3 Evaluation of the modified U-net for detecting foreign contaminants in terms of confusion matrix (contaminants detected are positive class) of training set, validation set and test set

Confusion matrix	True class		Validation set		Test set	
	Training set					
	Positive	Negative	Positive	Negative	Positive	Negative
Predicted class Positive	2000	0	741	18	1128	59
Negative	0	1000	59	382	72	541

Table 4 Precision, recall, F-score, and accuracy of training set, validation set and test set

	Training set	Validation set	Test set
Precision	1.0	0.98	0.95
Recall	1.0	0.93	0.94
F-Score	1.0	0.95	0.94
Accuracy (%)	100	94	93

to detect impurities in bread in this study. However, there are still some flaws during the process. Firstly, the cutout is not precise in the training set, which would interfere with the neural network. Unsupervised learning could be applied to explore whether it could segment the foreign contaminants. Secondly, the instinct property of foreign contaminants has an adverse effect on the U-net; the new solution is to train more samples. Thirdly, the number of samples which have been discussed before is limited by some condition. For the next step, more samples are required. Meanwhile, types of

Table 5 The accuracy of contaminants region segmentation in the test set

Foreign contaminants	Metallic iron	Polypropylene plastic	Hair	All
Sum	600	600	600	1800
Misclassified number	32	41	71	144
Accuracy (%)	95	93	92	92

foreign contaminants are needed as well to improve the generalization ability of the machine.

Conclusion

In summary, it can be demonstrated that foreign contaminants in bread could be detected based on NIR spectroscopy and CV techniques. They could be combined to apply further, which would make satisfactory results. For NIR spectroscopy technology, typical spectra of pure substances were comparatively analyzed due to the difference of distinct components, which reflect various vibrations in the near infrared range. In addition, classification effectivities of two modeling methods, distance match, and discriminant analysis are compared combined with diverse pre-processing methods, including Savitzky-Golay smoothing, the first derivative and the second derivative. It suggests that discriminant analysis with Savitzky-Golay smoothing has the

highest accuracy and F-score above all. On the other hand, the possibility of detecting foreign contaminants in bread with deep learning has been explored. A modified U-net was applied in the segmentation of foreign contaminants with excellent performance.

These results have proven the feasibility of these two technologies to detect foreign contaminants in bread, which is desired for the rapid non-destructive property for the food industry, beneficial to the quality control and accelerating safety risk assessment of food. What has been accomplished in this research work build a foundation for the development of new on-line detection methods of bread based on NIR spectroscopy and CV techniques.

Acknowledgements This work was supported by the National Key R&D Program of China (2017YFC1600805).

References

1. J. Quilez, J. Salas-Salvado, *Nutr. Rev.*, 2012, 70(11), 666–678 (2012)
2. F.N. Hyman et al., *Public Health Rep.* **108**(1), 54–59 (2009)
3. W. Zhang et al., *Crit. Rev. Food Sci. Nutr.* **58**(14), 2386–2397 (2017)
4. H.J. Shin et al., *Food Chem.* **245**, 282–288 (2018)
5. C. Wang et al., *Food Control* **97**, 100–104 (2019)
6. D. Wang et al., *Opt. Eng.* **57**(4), 043103 (2018)
7. M. Edwards, *Detecting Foreign Bodies in Food* (Woodhead Publishing Limited, Cambridge, 2004), pp. 86–171
8. D.G. Carvalho et al., *Food Chem.* **329**, 127142 (2020)
9. E.N. Wafula et al., *J. Food Eng.* **284**, 110056 (2020)
10. J.U. Porep et al., *Trends Food Sci. Technol.* **46**(2), 211–230 (2015)
11. M. Kawasaki et al., *Comput. Electron. Agric.* **63**(1), 22–27 (2008)
12. M. Lesteur et al., *Bioresour. Technol.* **102**(3), 2280–2288 (2011)
13. E.M. Achata, *Powder Technol.* **336**, 555–566 (2018)
14. R.A. Viscarra Rossel et al., *Geoderma* **131**(1–2), 59–75 (2006)
15. N. Prieto et al., *Meat Sci.* **83**(1), 96–103 (2009)
16. L.C. Carvalho et al., *Food Control* **106**, 106695 (2019)
17. R. Moschetti et al., *J. Food Eng.* **118**(1), 1–7 (2013)
18. P. Oliveri et al., *Talanta* **199**, 270–276 (2019)
19. R. Moschetti et al., *Postharvest Biol. Technol.* **99**(6), 58–62 (2015)
20. R. Moschetti et al., *Postharvest Biol. Technol.* **87**(2), 88–94 (2014)
21. A. Biancolillo et al., *Microchem. J.* **145**, 252–258 (2019)
22. T. Brosnan, D.W. Sun, *J. Food Eng.* **61**(1), 3–16 (2004)
23. H. Zareiforoush et al., *Food Eng. Rev.* **7**(3), 321–345 (2015)
24. N. Vélezrivera et al., *Food Bioprocess Technol.* **7**(4), 1183–1194 (2014)
25. A. Vidal et al., *Food Bioprocess Technol.* **6**(12), 3412–3419 (2013)
26. D. Mery et al., *Food Bioprocess Technol.* **6**(8), 2093–2108 (2013)
27. P.A. Belan et al., *Measurement* **46**(1), 177–184 (2013)
28. D. Rong et al., *Comput. Electron. Agric.* **162**, 1001–1010 (2019)
29. X. Zhou et al., *Food Chem.* **321**, 126503 (2020)
30. W.S. Cotrim et al., *J. Food Eng.* **277**, 10999616 (2020)
31. A. Taheri-Garavand et al., *J. Food Eng.* **278**, 109930 (2020)
32. J. Chen et al., *Comput. Electron. Agric.* **173**, 105393 (2020)
33. J. Sarmiento et al., *Talanta* **216**, 120937 (2020)
34. H.F. Yuan, et al., In *Proceedings of the Second Asian Near Infrared Spectroscopy Conference*, Shanghai, China, 2010
35. V.T. Gabriel, P.J. Schoenmakers, *Anal. Chem.* **78**(13), 4598–4608 (2006)
36. A. Savitzky, *Anal. Chem.* **36**, 1627–1639 (1964)
37. X. Lijuan et al., *J. Food Eng.* **82**, 395–401 (2007)
38. R.D. Maesschalck et al., *Chemometr. Intell. Lab Syst.* **50**(1), 1–18 (2001)
39. P. Moallem et al., *Inf. Process. Agric.* **4**(1), 33–40 (2017)
40. A. Krizhevsky et al., *Advances in Neural Information Processing Systems 25* (NIPS, Cambridge, 2012)
41. L. Li et al., *Neurocomputing* **392**, 277–295 (2020)
42. O. Ronneberger, et al., In *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer. pp. 234–241 (2015)
43. J. Workman, L. Weyer, *Practical Guide to Interpretive Near-Infrared Spectroscopy* (CRC Press, Boca Raton, 2007)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.